UNSOLVED PART OF SCAM - LAND DOCUMENT FORGEDY

EXECUTIVE OVERVIEW & LIVE DEPLOYMENT LINKS

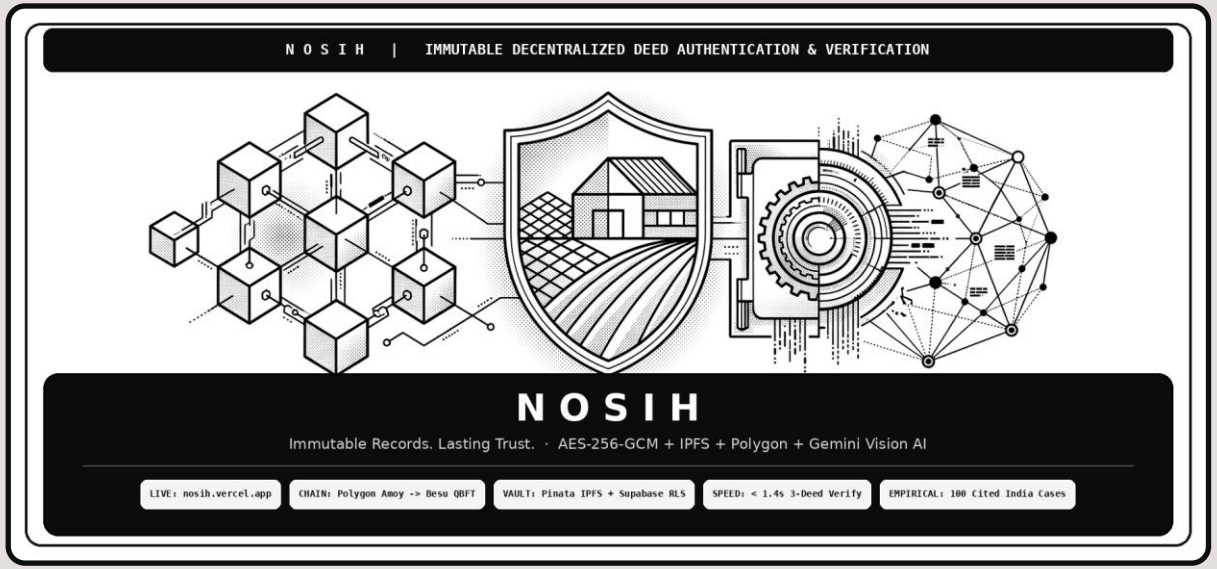
NOSIH

MISSION : Turning every registered land deed into a tamperproof immutable digital asset.

"Thirty years of savings. One small plot. One dream home. If a stranger claimed your home tomorrow with a forged deed, how fast could you prove it is yours?"

"FAKE IT UNTIL YOU MAKE IT — DIES HERE."

- **Live Web App :** nosih.vercel.app [Open App ↗]
- **Public Verifier :** nosih.vercel.app/verify [Verify Deed ↗]
- **Demo Video :** [Launch Video ↗]
- **GitHub Repository :** rmn2178/nosih-secure [View Code ↗]
- **Polygon Amoy Contract :** 0xE5a941d9...0952 [Polygonscan ↗]



< 1.4s	SHA-256	12 Fields	100 Cases
3-Deed Verify	IPFS Vault	On-Chain Proof	India Study

WORLD'S BIGGEST MOAT- LAND SCAM

What If the One Document You Trusted to Prove Your Home Is Yours Was Forged?

A STORY WE HEAR TOO OFTEN

Thirty years of savings.

One small plot.

One dream home.

The day Ravi went to lay the foundation, a stranger walked in holding a fake registered deed carrying the Ravi's own land he can't do anything because there is no quick and single trustable evidence to prove that the land is his.

Why It Happens Today:

- Paper stamps & seals are easily counterfeited
- Centralized portals trust a single admin login.
- Digilocker: Sub-Registrar Safe but not has cryptographical proof.

OF ALL PENDING CIVIL CASES IN INDIA

66%

Are land or property disputes are due to the land forgery and can't be resolved because there is no proof. [SC Ruling ↗]

PENDING STATE COURT CASES

2 in 3

Cases pending because of forged or manipulated land documents across 18 Indian States. [100 Cases PDF ↗]

GLOBAL DOCUMENT FRAUD

$12.3B

Up 37% are document scams— buyers have no instant way to verify a deed.

OUR SOLUTION — NOSIH

Combining Blockchain, IPFS Pinata & SHA-256 to Make Every Deed Self-Proving

1. BLOCKCHAIN (THE TRUST LAYER)

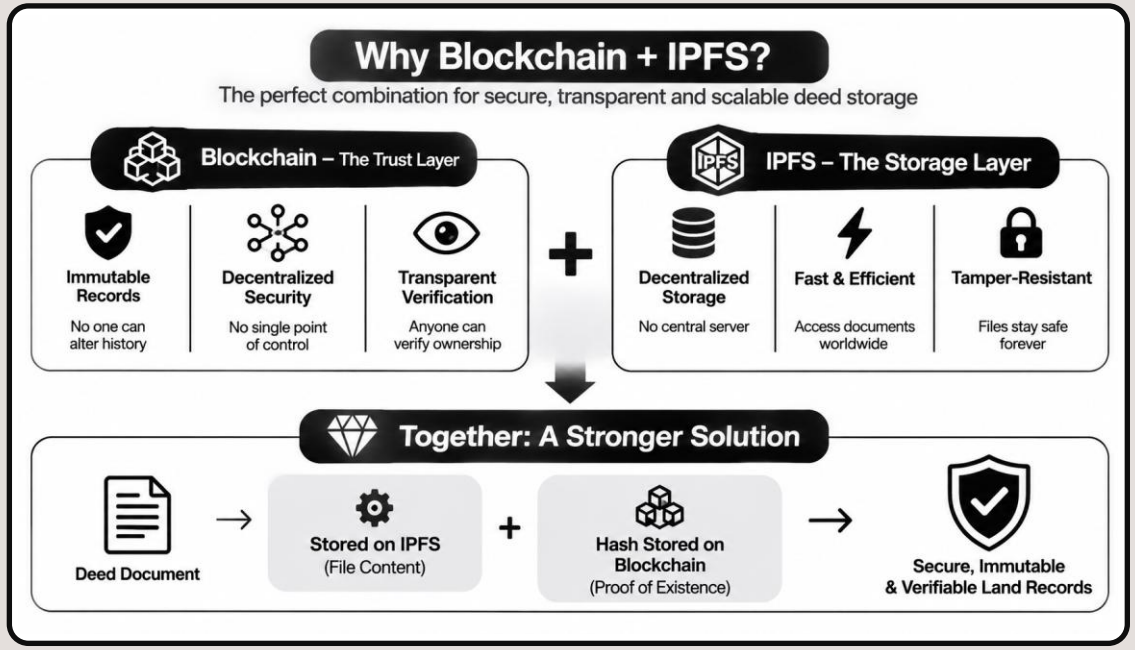
Stores data that can never be altered or modified — but storing a whole PDF document on-chain is far too costly.

2. IPFS PINATA (STORAGE LAYER)

Peer-to-peer distributed file system built for **scalable off-chain storage** & retrieval of large encrypted deed files by ID.

3. SHA-256 (DIGITAL FINGERPRINT)

Cryptographic hash function converting any document into a fixed **256-BIT** (32-byte) fingerprint stored on-chain with OCR metadata.

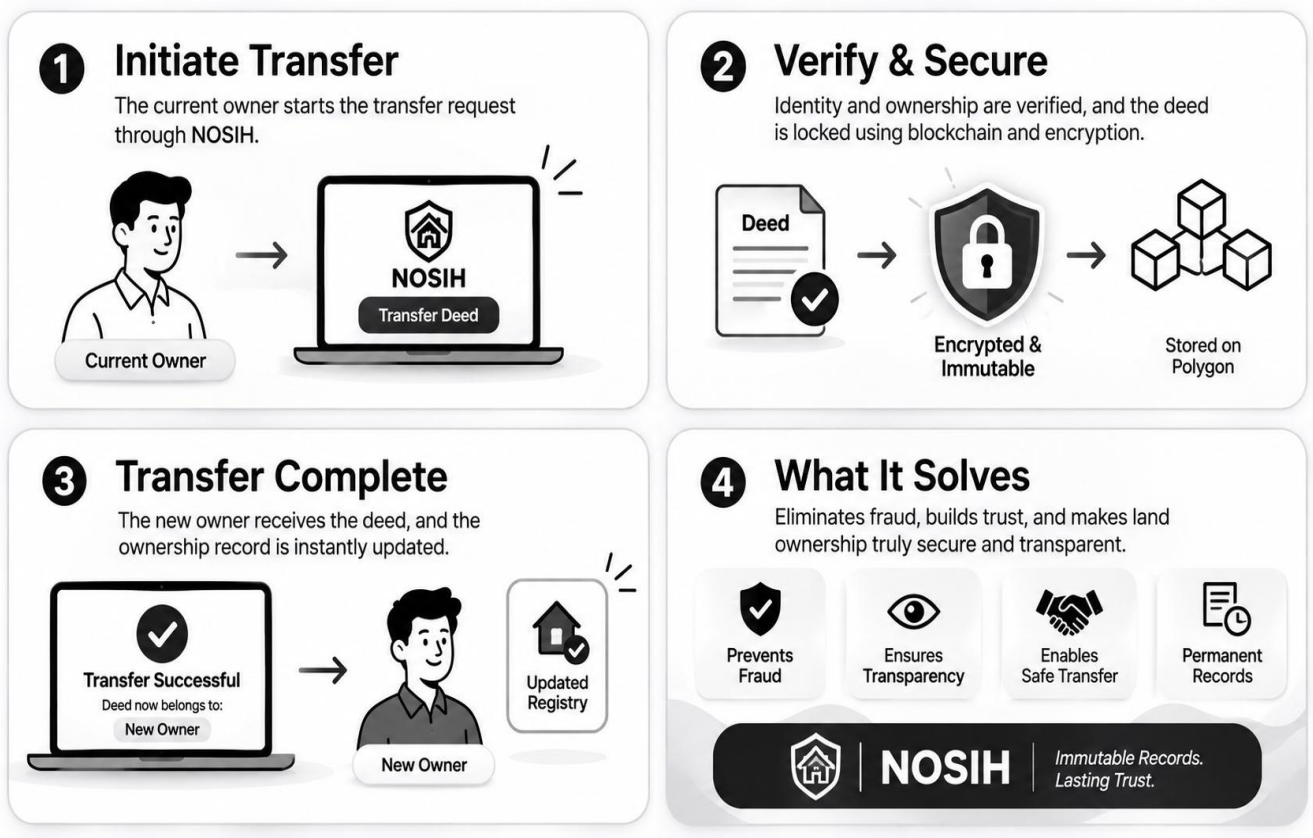


IMMUTABILITY + TAMPER-PROOF OWNERSHIP TRANSFER

- **The Immutability Paradox problem:** Blockchain is immutable, but land is not static — ownership transfers from person to person.
- **Turning It Into Our Biggest Plus:** When land transfers, we revoke the old active head and anchor the new deed with the Previous Owner's Hash locked inside the next owner's document.
- **Result:** The deed transfers cleanly while the entire previous-owner history stays 100% tamper-proof. [GitHub Architecture ↗]

MEET THE PRODUCT

"Fake It Until You Make It" Dies Here



1. AI-OCR METADATA EXTRACTION

Upload the deed PDF -> passed through Gemini OCR models to extract survey numbers, boundaries & metadata automatically.

2. SHA-256 + IPFS PINATA VAULT

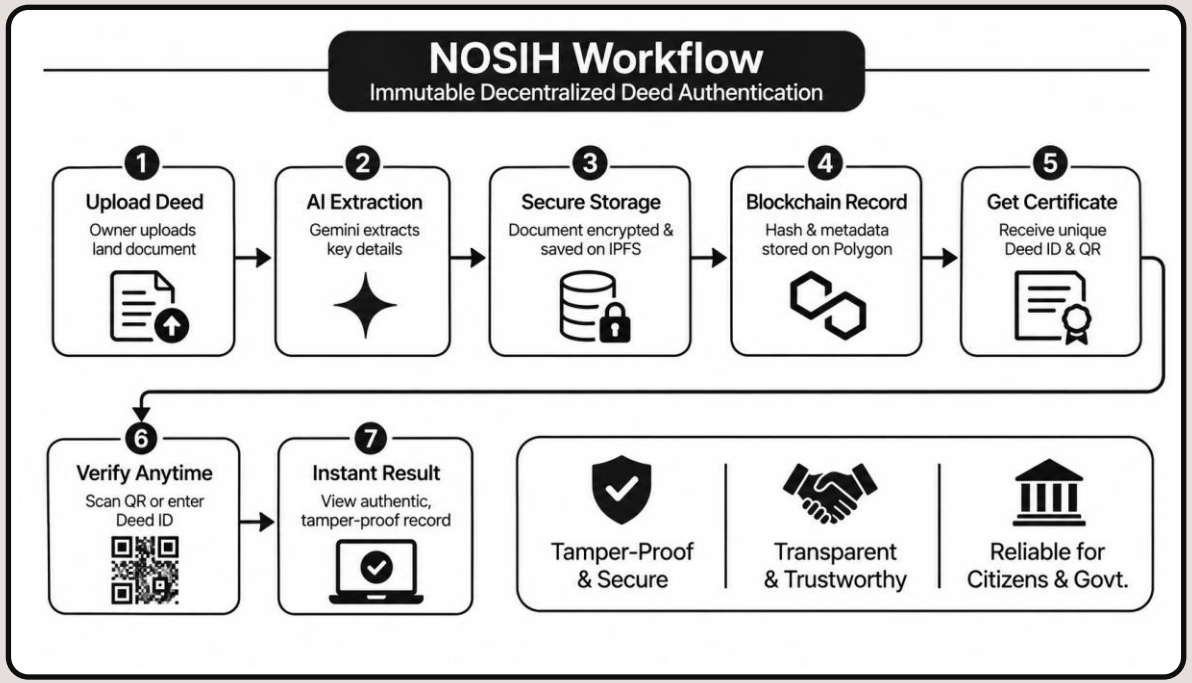
Takes the 256-bit SHA-256 fingerprint, stores the encrypted document on IPFS Pinata (by CID), and anchors metadata + hash on-chain.

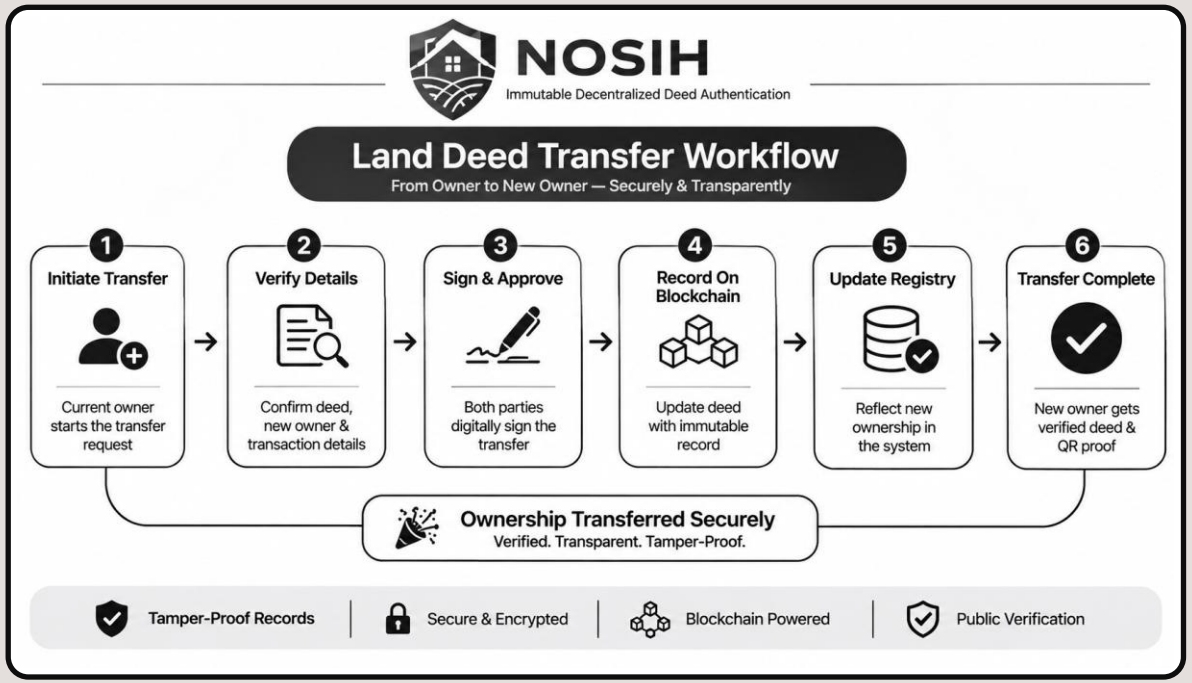
3. LAND ID & QR INSTANT

Enter the Land ID or scan the deed's QR code -> shows who owns the land and flags any 1-byte hash mismatch in 1.4 seconds.

HOW IT WORKS

7-Step Registration & Verification Pipeline · 6-Step Ownership Transfer Flow





STEP 1 · UPLOAD & OCR

Portal verifies 'PDF' - magic bytes -> Gemini OCR extracts 18 fields + computes 256-bit SHA-256 hash.

STEP 2 · IPFS + BLOCKCHAIN

AES-256-GCM encrypted PDF pinned to IPFS Pinata

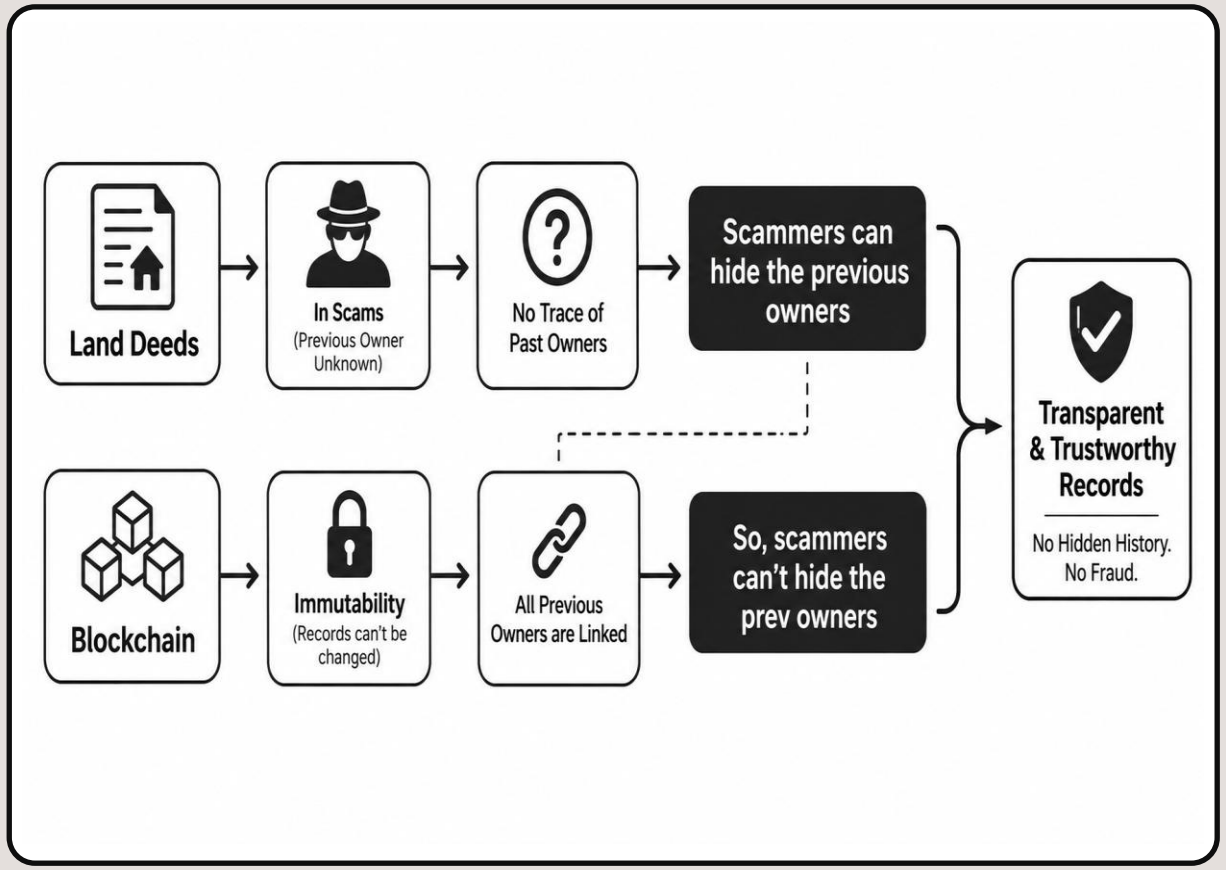
12 fields + previous-owner hash anchored on-chain [BLOCKCHAIN].

STEP 3 · SCAN QR OR LAND ID

Entering Land ID or scanning QR runs a live 3-way check (DataBase == On-Chain-Hash == upd-PDF)

KEY INNOVATIONS

Four Architectural Breakthroughs Built



1. KEYED PARCEL IDENTITY

Survey numbers are small and guessable.

Keyed HMAC-SHA256 stops brute-force enumeration on-chain while deriving deterministic LAND-XXXXXXXXXX IDs.

2. CONTINUOUS OWNERSHIP CHAIN

Every deed links previousRecordId & previousOwnerId

(A -> B -> C -> D)

Revoking a previous head restores the prior active owner as head.

3. THREE-WAY LIVE VERIFICATION

Every read 3 axes

(documentHash == propertyId == revoked)

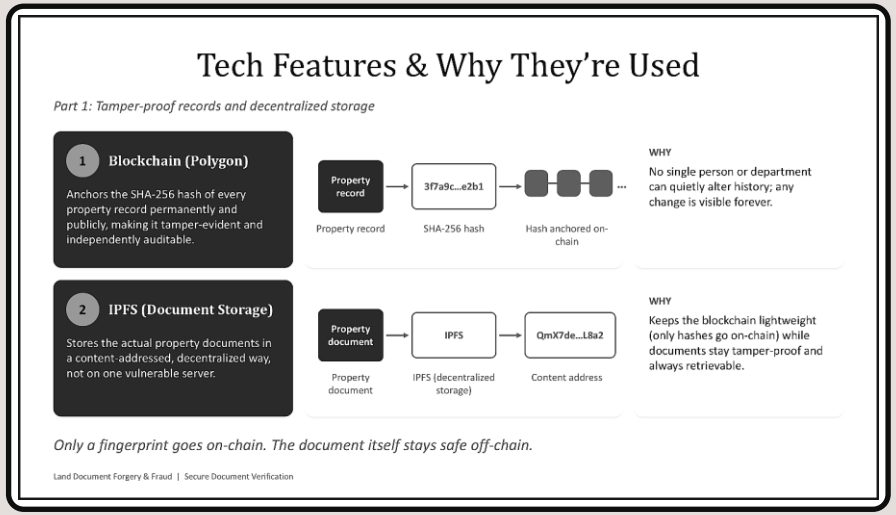
Even a 1-byte PDF edit fails.

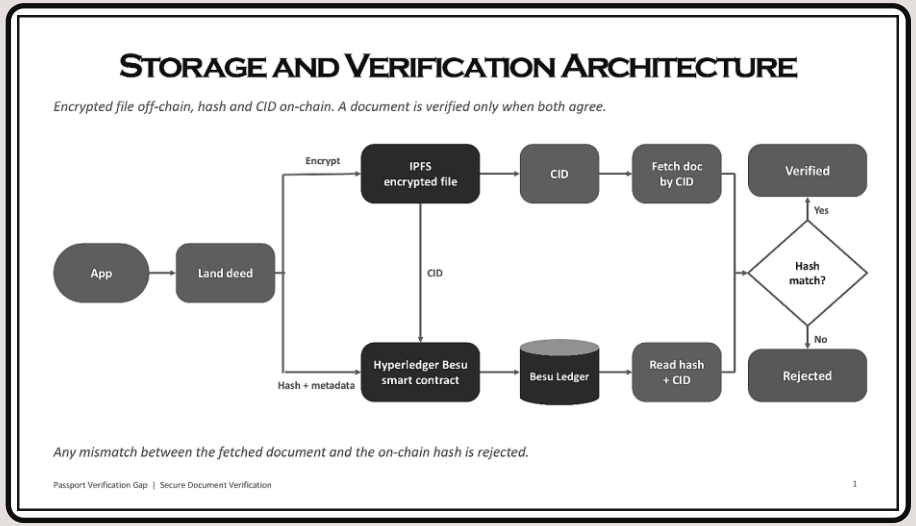
4. DUAL-PATH AI THAT PRE-FILLS

Automatically probes PDF text layers Vision OCR

reducing typing and human errors while keeping human registrars in control.

Cryptographic Core, Smart Contract & Decentralized Storage Layer





1. POLYGON AMOY (80002) → HYPERLEDGER BESU QBFT

WHY: Anchors every registration in 1 transaction (~4.8s); all verification reads are free view calls. Ready for 5-validator zero-gas Besu. [Explorer ↗]

2. SOLIDITY SMART CONTRACT

WHY : Stores 12 fields per PropertyRecord
Signed contract on blockchain.

3. PINATA IPFS + 5-GATEWAY RACE

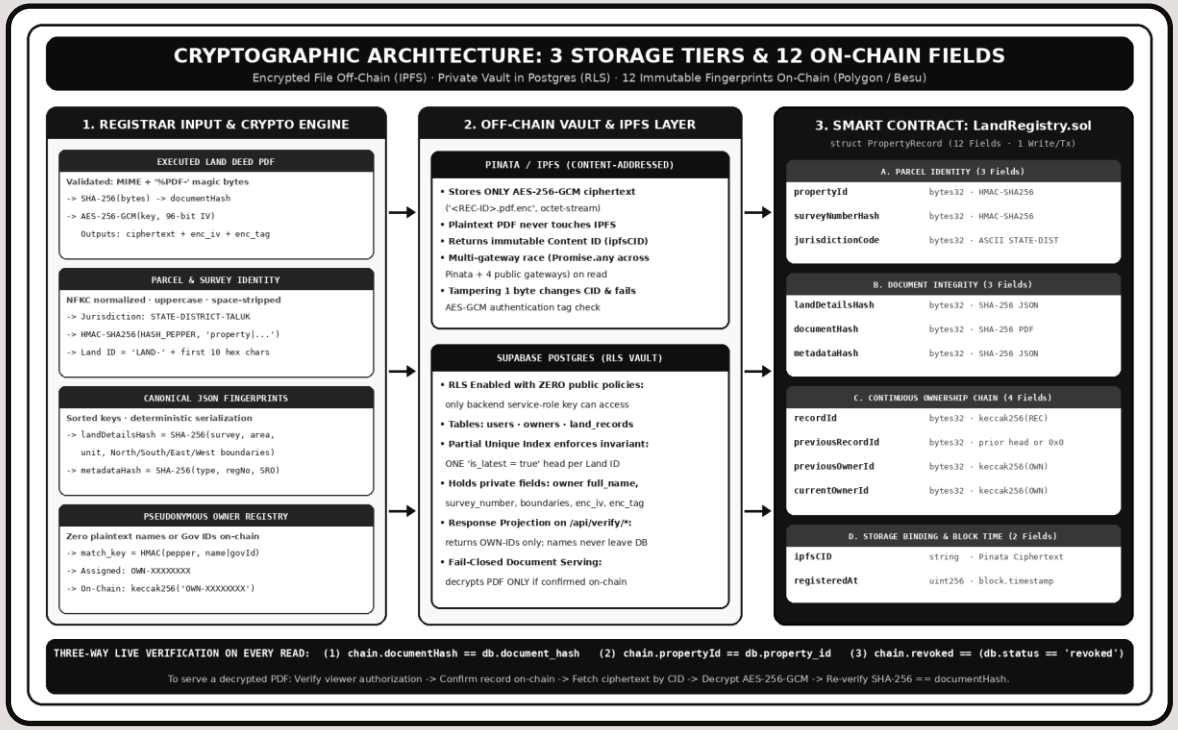
WHY: Pushes AES-256-GCM encrypted deeds off-chain IPFS + 4 public gateways via Promise.any on retrieval.

4. THREE HASH FAMILIES + AES-256-GCM

WHY: HMAC-SHA256 (parcel/survey privacy)
SHA-256 (PDF & canonical JSON fingerprints)
keccak256 (IDs), and AES-256-GCM (96-bit IV + auth tag).

BUILT WITH · PART 2

Gemini Vision AI, Relational Vault (RLS) & 14 Self-Verifying Preflight Checks



Click the image for the clearer view

1. GOOGLE GEMINI 3.6 FLASH + PDF.JS

WHY: Extracts 18 deed fields from digital PDFs or scanned pages in ~3.2s.

2. SUPABASE POSTGRES VAULT

WHY: Row-Level Security with 0 public policies guards owner names & IV/tags; partial unique index enforces 1 active head per Land ID.

3. NODE.JS 22 + EXPRESS 4 + REACT

WHY: Single-origin deployment not a complex deployment the whole complexity lies in the backend.

4. 14 AUTOMATED CHECKS

WHY: Built-in immune system catches drift before deploy. So no worries while deploying. Not just an prototype this was a product with ultimate test suite

WHO NEEDS THIS?

Where Trust Is the Bottleneck, Mathematical Proof Is the Product

OF INDIA'S CIVIL CASES

~66%

Are land or property disputes (1.1 Crore pending cases)

INVESTMENT THREATENED

$200B+

Delayed by land conflicts (20-year average dispute life)

ULPIN LAND PARCELS

40.58 Cr

Digitized parcels needing tamper-evident deed verification

1. STATE REGISTRATION DEPARTMENTS

Eliminates fake parent deeds, forged pattas.
insider portal overrides right at the registration counter.

2. BANKS & MORTGAGE LENDERS

Instant < 1.4s title and ownership-chain verification before sanctioning property loans

So the best investment that they can make is this NOSIH

3. HOMEBUYERS,FARMERS & NRIs

Scan the QR code or enter the Land ID to verify the real owner and document originality before paying.

4. COURTS & LEGAL TEAMS

Tamper-proof historical ownership chain (previous-owner hashes locked block-to-block) cuts years of litigation.

NOSIH

FUTURE PLAN & LIVE ACCESS

Private Permissioned Blockchain Layer for Personal Data & State-Wide Rollout

KEY REAL-WORLD CONSTRAINT: PRIVATE BLOCKCHAIN LAYER OVER PUBLIC CHAIN

- **Why a Private Layer Is Essential:** Citizen personal data cannot sit on a public blockchain. Our institutional architecture deploys a **private 5 layer validator best network.**
- **which can be deployed by the government of the country then maintained.**

PHASE 1 · LIVE MVP (DONE)

Months 1-2

Polygon Amoy + Pinata IPFS ·

₹0-₹3,500/month

[Live App ↗]

PHASE 2 · PRIVATE BESU PILOT

Months 3-8

5 SROs · 5-Node Besu + 2-of-2 Co-Sign

5,000 Deeds (₹8.5L) [Repo ↗]

PHASE 3 · STATE ROLLOUT

Months 9-12 ·

NIC Vishvasya Cloud · Zero Gas · <350ms Verify

[Architecture ↗]

CLICK OR SCAN TO TEST NOSIH LIVE

- **Live Web App :** https://nosih.vercel.app/ [Launch ↗]
- **Public Verifier :** https://nosih.vercel.app/verify [Verify ↗]
- **39-Step Guided Tour :** https://nosih.vercel.app/?tour=1 [Tour ↗]
- **GitHub & Case Study :** rmn2178/nosih-secure [100 Cases PDF ↗]

